



Sales Data Modernization & Governance Architecture

ORACLE, TINA, SSIS & SQL SERVER
→ GOOGLE CLOUD PLATFORM

OBJECTIVE

Modernize the legacy Oracle, Tina, SSIS, and SQL Server ecosystem by migrating to a scalable Google Cloud Platform architecture that improves data quality, governance, customer hierarchy management, reporting capabilities, and customer master data management.

Business Impact:

- This initiative establishes a trusted customer data foundation by combining cloud modernization with data governance and master data management.
- The result is improved data quality, stronger hierarchy and territory management, automated governance controls, and scalable reporting capabilities for future growth.

KEY GOALS

- *Eliminate SSIS and SQL Server dependencies*
- *Centralize customer master data*
- *Create a trusted Golden Customer Record through entity resolution, survivorship rules, and third-party enrichment.*
- *Introduce governance checkpoints during ingestion*
- *Automate duplicate detection and CLE validation*
- *Integrate third-party enrichment sources*
- *Enable near real-time reporting and data quality monitoring*
- *Support future Salesforce and Dynamics integrations*

Challenges:

- Duplicate Customers
- Inconsistent CLEs
- Limited Lineage
- Manual Governance
- Reporting Delays
- Territory Assignment Risks

Current State Architecture



Problems:

- Nightly batch only
- Limited governance
- Duplicate customers
- Manual hierarchy maintenance
- Limited lineage visibility
- Reporting delays

Target State GCP Architecture



1. Cloud Data Fusion

- Replacement for SSIS extraction

2. Cloud Storage

- Landing zone

3. BigQuery

- Enterprise data warehouse

4. Dataflow

- Transformation engine

5. Looker

- Reporting

6. Cloud Composer

- Scheduling & orchestration

Target State GCP Architecture Reasoning

Source data from Oracle and the Tina application is ingested through Cloud Data Fusion and stored in a raw landing zone. Dataflow then performs transformations before records enter the Data Governance and MDM layer, where CLE validation, duplicate detection, entity resolution, hierarchy validation, and enrichment are applied.

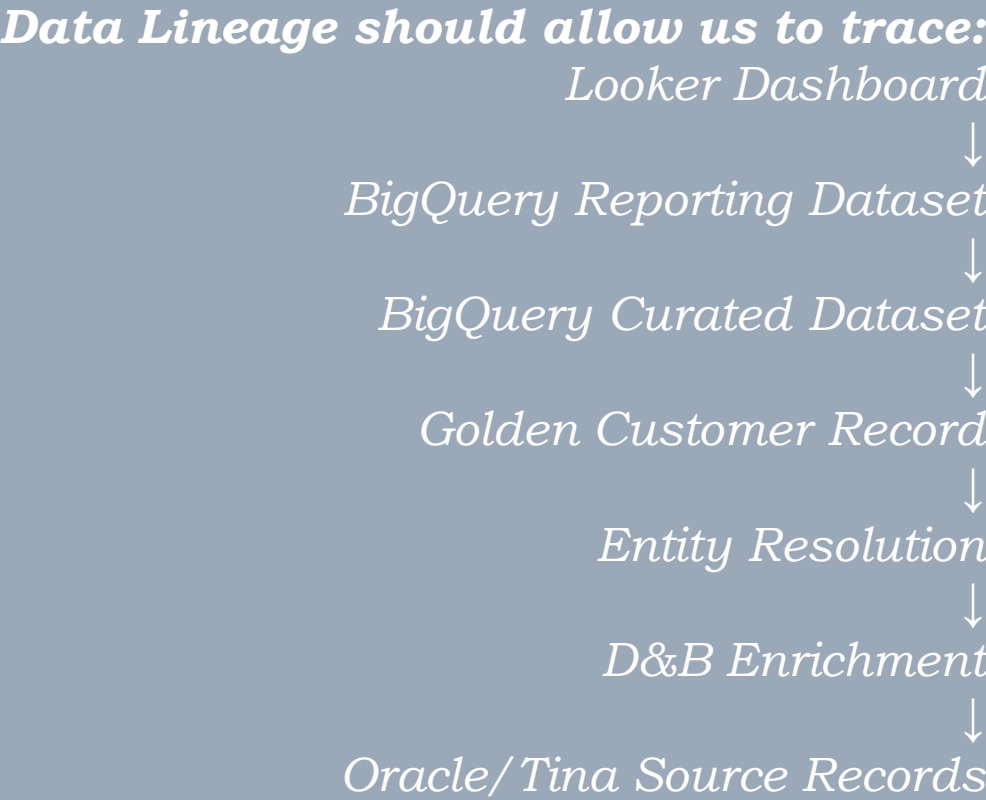
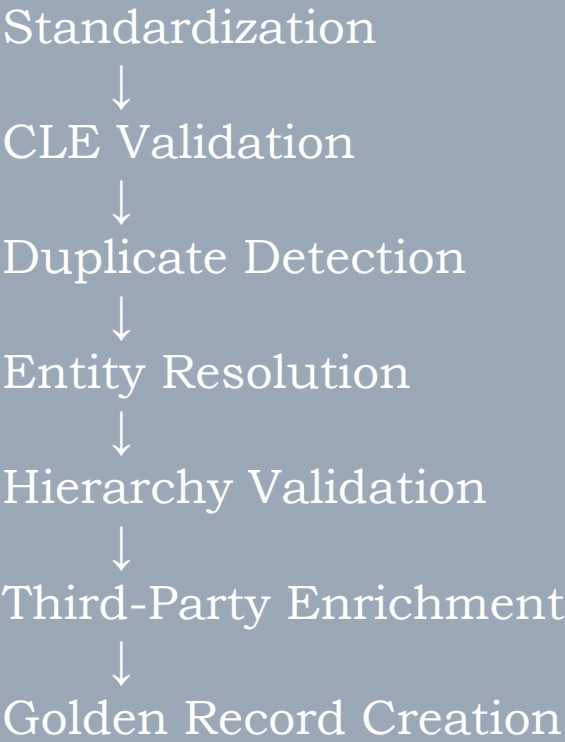
These processes produce a trusted Golden Customer Record, which serves as the single source of truth for customer data. The curated and reporting datasets in BigQuery support Looker dashboards, while Cloud Composer orchestrates and monitors the entire workflow.

- I have used Cloud Data Fusion as the ingestion and integration layer because it provides a low-code approach for connecting Oracle and legacy systems and serves as a natural replacement for SSIS extraction workflows.
- I used Dataflow as the transformation and processing layer because it can efficiently handle data standardization, CLE validation, duplicate detection, entity resolution, hierarchy validation, and third-party enrichment at scale.
- Together, they separate ingestion from governance and transformation responsibilities, resulting in a more scalable and maintainable architecture.

- D&B and ZoomInfo are used as external enrichment sources within the Data Governance and MDM layer.
- After customer records are standardized and validated, the pipeline queries these providers to verify legal entities, confirm parent-child hierarchies, identify duplicates using DUNS numbers and company attributes, and enrich customer records with firmographic information such as revenue, employee counts, industry classifications, and ultimate parent relationships. This enrichment process helps create a trusted Golden Customer Record before the data is promoted to the curated reporting layer.

Data Lineage & Governance Plan

All transformations and validation decisions are captured through data lineage to ensure traceability and auditability.



1. CLE Validation	2. Duplicate Detection	3. Entity Resolution	4. Hierarchy Validation
- Acme corp - Acme corporation - Acme corp llc Normalize & Create Master Record	- Exact match - Fuzzy match - Tax ID match - DUNS match	- Exact Match - Fuzzy Match - D&B / ZoomInfo Verification - Survivorship Rules - Hierarchy Resolution	- Parent-Child Validation - Third-Party Verification (D&B/ ZoomInfo) - Ultimate Parent Resolution - Circular Relationship Detection - Hierarchy Completeness Checks

Migration Runbook

Phase 1: Discovery

- Inventory Oracle tables
- Inventory Tina datasets
- Document SSIS packages

Phase 2: Assessment

- Data profiling
- Duplicate analysis
- Hierarchy assessment

Phase 3: Build Target Environment

- GCS
- BigQuery
- Dataflow
- Composer

Phase 4: Historical Migration

Load:

- 5 years
- 10 years
- Full history into BigQuery

Phase 5: Validation

Perform:

- Record Count Validation
- Aggregate Validation
- Hierarchy Validation
- CLE Validation
- Business Rule Validation

Phase 5.5: User Acceptance Testing (UAT)

- Business Validation
- Sales Validation
- Governance Validation

Phase 6: Parallel Run

Compare outputs from:

- SQL Server
- BigQuery

Phase 7: Cutover

- Move production reporting to GCP.

Phase 8: Hypercare

- 30-day monitoring period.

Reporting Mockup

Executive Dashboard will include metrics, like -

1. Sales Account Coverage

- Active Accounts: 42,000
- Validated CLEs: 41,250

2. Data Quality Metrics

- Duplicate Rate: 1.8%
- Completeness Score: 96%
- Hierarchy Coverage: 98%

3. Golden Record Coverage

- Golden Record Coverage: 97%

3. Governance Metrics

- Missing Tax IDs
- Missing Parent Company
- Unverified Entities
- Failed Enrichment Records

4. Territory Conflict View

- Account
- Owner
- Parent Company
- Conflict Status

4. Territory Conflict Count

- Open Territory Conflicts: 12
- Closed Territory Conflicts: 7

- This executive dashboard provides visibility into the overall health, quality, and governance of customer and sales account data. It tracks key metrics such as active accounts, validated customer legal entities (CLEs), duplicate rates, data completeness, and hierarchy coverage.
- The dashboard also highlights governance issues including missing tax IDs, unverified entities, and failed enrichment records, while providing a territory conflict view to support accurate account ownership and territory assignment. Together, these metrics help ensure trusted customer data, stronger governance, and improved sales operations.

Business Benefits

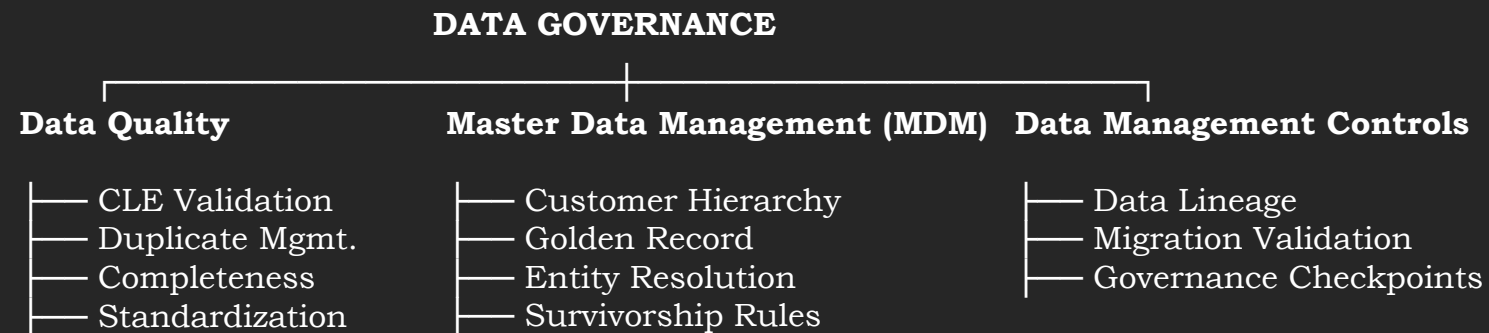
Expected Outcomes:

- Modern cloud-native architecture
- Trusted Golden Customer Records and improved customer master data quality
- Reduced duplicate customer records
- Better visibility into corporate hierarchies
- Automated governance controls
- Faster onboarding of sales accounts
- Increased confidence in territory assignment
- Scalable foundation for future Salesforce and Dynamics initiatives

- The proposed solution modernizes the existing data ecosystem by migrating it to a scalable, cloud-native GCP architecture while strengthening data governance and customer data quality. Automated validation, duplicate management, and hierarchy resolution will improve the accuracy of customer records, reduce territory conflicts, and streamline sales account onboarding.
- The platform will provide trusted reporting, better visibility into customer hierarchies, and a scalable foundation for future Salesforce and Microsoft Dynamics initiatives.

Data Governance Perspective

While this is a cloud migration initiative, I view it equally as a data governance and master data management initiative. The GCP services provide the technology foundation, but the real business value comes from implementing governance controls such as CLE validation, duplicate management, hierarchy management, data lineage, and migration validation to create a trusted customer master.



Components of an Enterprise Data Governance Framework.

1. CLE VALIDATION (Customer Legal Entity Validation)

CLE Validation is the process of verifying that a customer account represents a legitimate, unique, and legally recognized business entity.

2. DUPLICATE MANAGEMENT

Duplicate Management identifies, prevents, and resolves multiple records that represent the same customer through entity resolution techniques.

3. CUSTOMER HIERARCHIES

Customer Hierarchies define relationships between parent companies, subsidiaries, divisions, and branches.

4. DATA GOVERNANCE CHECKPOINTS

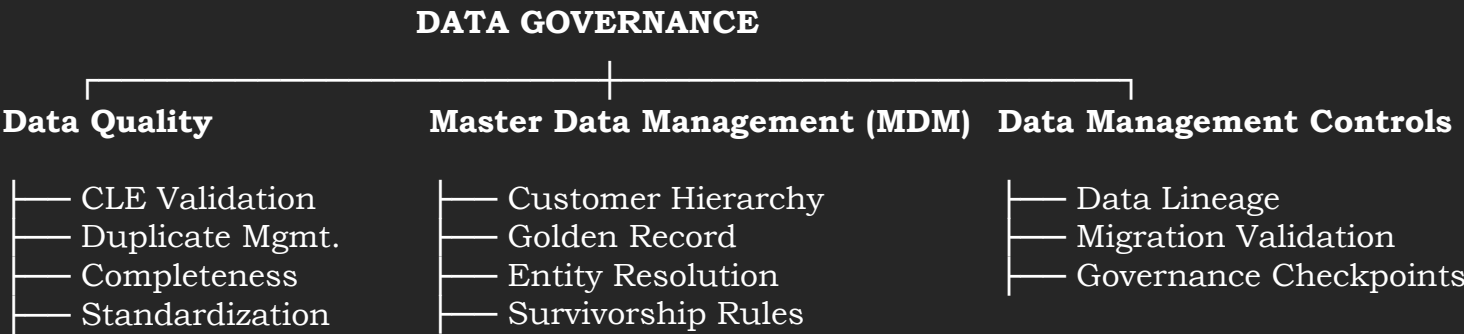
Governance checkpoints are control points within the pipeline where data quality and business rules are evaluated.

5. MIGRATION VALIDATION

Migration Validation ensures data migrated from the legacy environment matches the source system.

6. DATA LINEAGE

Data Lineage tracks where data came from, how it was transformed, and where it is consumed.

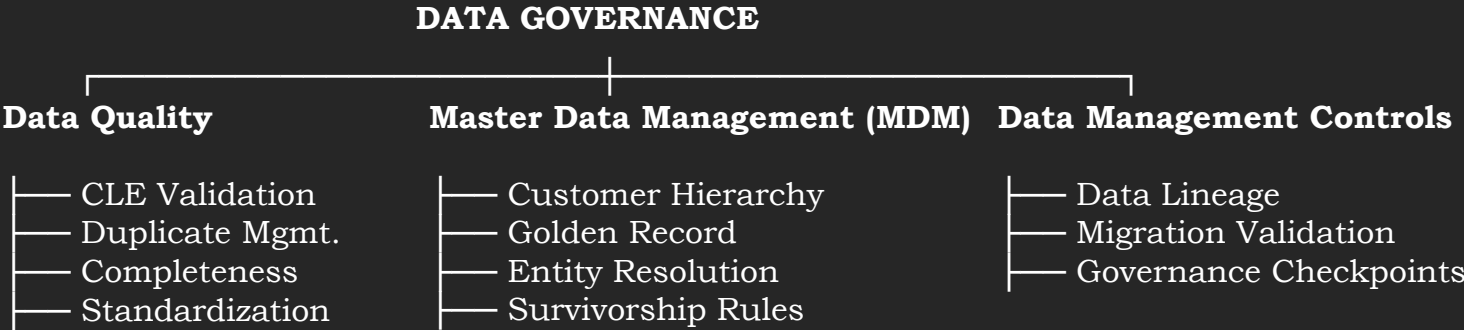


How the Six Components Fit Together

Our objective of the solution –

To create a trusted customer master by validating legal entities, eliminating duplicates, managing customer hierarchies, enforcing governance checkpoints, maintaining full data lineage, and validating migrated data before it is consumed by reporting and sales operations.

- Oracle + Tina
- ↓
- Data Ingestion
- ↓
- Standardization
- ↓
- CLE Validation
- ↓
- Duplicate Detection
- ↓
- Entity Resolution
- ↓
- Hierarchy Validation
- ↓
- Enrichment
- ↓
- Golden Record Creation
- ↓
- BigQuery Curated Layer
- ↓
- Migration Validation
- ↓
- Looker Reporting



Strategic Vision & Business Value

While this initiative modernizes the legacy Oracle, Tina, SSIS, and SQL Server ecosystem onto Google Cloud Platform, I view the larger opportunity as establishing a governed Customer Master Data platform.

Through entity resolution, CLE validation, hierarchy management, enrichment, and lineage controls, the proposed architecture creates trusted Golden Customer Records that improve territory management, reporting accuracy, and sales effectiveness while providing a scalable foundation for future Salesforce and Dynamics initiatives.

Thank you!